Spotter

Project Charter and Team Charter

Version 1.0, October 2026. Prepared alongside SRS v2.0.0. Part 1 is the Project Charter; Part 2 is the Team Charter.

PART 1: PROJECT CHARTER

1.1 Project Overview

Item	Detail
Project name	Spotter
Description	A cross-platform (iOS, Android, web) parking application that unifies commercial parking booking (Layer 1) with a peer-to-peer driveway marketplace prototype (Layer 2).
Project manager	Isabelle Cruz
Product owner	Daniel Okafor
Project sponsors	Sponsor group
Start date	September 10, 2026
Target completion	December 17, 2026: a 2-week planning phase followed by six 2-week sprints (14 weeks)
Approved budget	Unlimited; ballpark $427,561 (Base + 15% Contingency)
Team	12 people, 9.0 full-time equivalents
Delivery approach	Scrum with 2-week sprints, tracked in Jira (86 tickets across 10 epics)
Governing document	Software Requirements Specification (SRS) v2.0.0: 17 functional requirements, 13 non-functional requirements, 29 use cases

1.2 Purpose and Business Justification

Drivers spend significant time searching for parking, which causes congestion, frustration and lost productivity. Parking operators lack the tools to monitor occupancy, manage pricing and improve utilization of their facilities.

Spotter gives drivers one app to find, reserve and pay for parking in real time, and gives operators a web dashboard for occupancy, revenue analytics and pricing. Commercial booking apps and driveway-sharing apps largely exist separately today. Combining them in a single search experience is the product's differentiator, and the Layer 2 prototype lets sponsors evaluate the driveway marketplace before committing to build it.

1.3 Objectives

- Deliver Layer 1 on iOS, Android and web, covering the core booking requirements (FR1-FR9) and the operator dashboard requirements (FR13-FR17), by December 17, 2026.
- Deliver Layer 2 as high-fidelity mockups and written user-flow documentation (FR10-FR12), with no live backend.

- Meet the performance and reliability targets: search results within 2 seconds (NFR2), 99% uptime for core booking during operating hours (NFR5), and operator occupancy data no more than 1 minute old (NFR11).
- Meet the security requirements: PCI-DSS compliant payment handling (NFR4) and strict separation of operators' data (NFR13).
- Complete the project within the approved budget and timeline.
- Pass operator user acceptance testing and the final sponsor demo.

1.4 Scope

In scope: Layer 1, Commercial Parking Booking (fully functional).

- Registration, login, password reset and vehicle profile management
- Real-time map search and price comparison across garages, lots, event/venue, airport and metered/on-street parking
- Reservation and digital payment, QR/barcode digital passes, and session management (view, extend, cancel)
- Push and email notifications, reservation history, receipts and saved locations
- Operator dashboard (occupancy, analytics, pricing, facility settings, report export) and an admin console

In scope: Layer 2, Peer-to-Peer Driveway Marketplace (prototype only).

- Mockups for host listing creation, driveway search results in the main map view, booking confirmation and a two-sided rating display
- Written user-flow documentation for hosts and drivers. No functional backend, payments, host verification or insurance handling.

Out of scope.

- Full host onboarding, identity verification and payouts; insurance, liability and dispute-resolution workflows
- Business and fleet parking accounts; in-car integrations (CarPlay, Android Auto)
- AI-based support or chat; real-time IoT sensor integration; offline mode

1.5 Key Deliverables

Deliverable	Description	Timing
Requirements baseline	SRS with requirements, use cases and project management sections, versioned	Ongoing, baselined in week 2
Layer 1 applications	iOS, Android and web clients, operator dashboard, admin console	Sprints 1-5, hardened in Sprint 6
Backend and infrastructure	APIs, data layer, integrations, CI/CD, environments, logging and monitoring	Sprints 1-5
Layer 2 prototype	Mockups and user-flow documentation	Sprints 4-5
Quality evidence	Test plan, system test results, security and PCI-DSS review, load test results	Sprints 1, 4, 5, 6
Management artifacts	WBS, Gantt chart, Jira backlog, risk register, communication plan, cost plan, RACI	Maintained every sprint
Release package	Production release, final documentation and final presentation	Sprint 6

1.6 Milestones

ID	Milestone	Week	Target date
M1	SRS baselined and planning complete	2	Sep 23, 2026
M2	Sprint 1 review complete	4	Oct 8, 2026
M3	Mid-project sponsor demo (Sprint 3 review)	8	Nov 5, 2026
M4	Feature complete and code freeze (Sprint 5 end)	12	Dec 3, 2026
M5	Final sponsor presentation	14	Dec 17, 2026

Milestone weeks follow Gantt Chart v1.1; M1 to M5 are proposed and confirmed at sponsor sign-off.

1.7 Budget Summary

Item	Amount	Basis
Labor (12 roles, 5,040 hours)	$362,292	Allocation percentage x 560 hours x hourly rate (Resource and Cost Plan)
Non-labor	$9,500	Cloud hosting $3,500, mapping API $1,500, payment sandbox and compliance tooling $2,000, tool licenses $2,500
Base cost	$371,792	Labor plus non-labor
Contingency reserve	$55,769	15% of base cost, tied to the highest-scoring risks (S1, T1, T4). Released only with sponsor approval.
Total approved budget	**$427,561**	Maximum authorized spend

1.8 Constraints and Assumptions

Constraints

- Fixed, semester-long timeline of 14 weeks
- PCI-DSS and data security and privacy compliance for payment and personal data
- Dependence on third-party mapping, payment and push notification services
- Consistent functionality across iOS, Android and web, and an always-on network connection
- Backend must scale for demand spikes around events; Layer 2 is limited to prototype fidelity

Assumptions

- Funding is available up to the approved $427,561, and the team allocations in Part 2 are honored.
- Third-party mapping and payment services stay available and within their published API terms.
- Layer 2 stays at prototype fidelity, and zoning and HOA restrictions on driveway rental are a future-phase concern.
- Sponsors and operator representatives are available for the scheduled demos and acceptance testing.

1.9 Key Risks

Top-scoring risks from the risk register (score = probability x impact, maximum 25). The full register has 20 risks across technical, schedule, financial and people categories.

ID	Risk	Score	Response
S1	Fixed 14-week timeline has no slack, so early delays compound across dependent tasks	16	Mitigate: track the critical path weekly and flag slippage early

ID	Risk	Score	Response
T1	Third-party API failure or rate limiting disrupts core booking	12	Mitigate: graceful fallbacks and error handling (NFR7); monitor provider status
S2	Frontend features blocked by incomplete backend services	12	Mitigate: prioritize backend services ahead of dependent UI tickets
S3	Testing and QA window is short and likely to be compressed	12	Mitigate: hard checkpoint dates; de-scope lower-priority tickets if needed
P1	Uneven skill distribution creates single points of failure	12	Mitigate: pair less-experienced members with component owners
P4	Critical system knowledge concentrated in one person	12	Mitigate: code documentation and knowledge-sharing sessions
T4	Payment or personal data handling fails PCI-DSS requirements	10	Avoid: established payment service abstraction and a PCI-DSS checklist

1.10 Stakeholders

Power / interest	Stakeholders	Approach
High power, high interest: manage closely	Project sponsors; parking facility operators; project leadership (PM, PO); development teams	Regular consultation, demos and joint decisions
High power, low interest: keep satisfied	City transportation departments; finance and billing personnel; payment and mapping providers	Scheduled briefings and reports; monitor dependencies
Low power, high interest: keep informed	Drivers and vehicle owners; mobile app users; system administrators; subject matter expert	In-app updates, release notes, feedback channels, bi-weekly admin syncs
Low power, low interest: monitor	Homeowner hosts (prototype only); guest users; push notification platforms	Minimal effort; revisit when Layer 2 goes live

1.11 Authority and Governance

Decision	Decision maker	Notes
Backlog priority and acceptance of completed work	Product Owner	Accepts tickets that meet their acceptance criteria
Scope changes (adding or removing requirements)	Product Owner proposes; sponsors approve	Sponsor approval required when budget or timeline is affected
Schedule and resource adjustments within allocations	Project Manager	Reflected in the Gantt chart and communicated at the next sprint review
Release of contingency funds	Sponsors, on PM request	Logged in the cost plan with the risk it addresses
Architecture and technical standards	Backend/API lead with team leads	Per the RACI matrix
Release go/no-go	Project Manager	With input from the PO and QA

Escalation path: team member, then team lead, then Project Manager, then sponsors. Issues that threaten a milestone or the budget go to the PM immediately.

1.12 Success Criteria

- All Layer 1 requirements are implemented, tested and accepted by the Product Owner, and the Layer 2 mockups and user flows are accepted.
- Non-functional targets are met in testing, including search under 2 seconds and a clean PCI-DSS review.
- No critical or high-severity defects remain open at release.
- The project finishes by December 17, 2026 and within $427,561.
- Operators pass user acceptance testing and sponsors sign off at the final demo.

1.13 Approval

By signing, the approvers authorize the project to proceed under the scope, budget and timeline above.

Role	Name	Signature	Date
Project sponsor			
Project sponsor			
Project manager	Isabelle Cruz		
Product owner	Daniel Okafor		

PART 2: TEAM CHARTER

2.1 Mission

We build Spotter so drivers can find, reserve and pay for parking in seconds and operators can see and manage their facilities in real time. We deliver it on schedule, secure, and consistent across iOS, Android and web, and we look after each other and the quality of our work while we do it.

2.2 Team Scope

The team is responsible for:

- Requirements, design, development, testing and release of Layer 1
- The Layer 2 mockups and user-flow documentation
- Keeping the SRS, WBS, Gantt chart, Jira backlog, risk register, cost plan and communication plan current

Outside the team's scope:

- Operating the live service after release, and building Layer 2 beyond prototype
- The internals of third-party providers, and anything listed as out of scope in the Project Charter

2.3 Goals

- Ship a working Layer 1 on all three platforms by December 17, 2026.
- Complete at least 85% of committed tickets in every sprint, and keep milestones M1 to M5 on schedule.
- Release with no open critical or high-severity defects and a clean PCI-DSS review.
- Make sure every critical service has a second person who can maintain it, reducing single points of failure.
- Keep project documents in sync: update the SRS, WBS, Gantt, Jira and risk register at each sprint close.
- Run honest retrospectives and act on what we learn.

2.4 Strengths and Skills

Area	Team members	Strengths
Delivery and product leadership	Isabelle Cruz (Senior), Daniel Okafor (Senior)	Scheduling, risk and sponsor management; backlog ownership and acceptance
Requirements and design	Priya Nair (Mid), Lucas Meyer (Mid)	SRS, use cases and traceability; wireframes, user flows and Layer 2 mockups
Domain expertise	Rosa Delgado (Senior)	Parking operations and municipal parking regulations; validates pricing, occupancy and metered-parking logic
Backend and data	Carlos Vega (Senior)	Reservation, search and inventory services; data layer; role-based access control
Payments and compliance	Ibrahim Yusuf (Senior)	Payment flow, refunds, receipts and PCI-DSS checkpoints
Client applications	Aisha Rahman (Mid), Tom Nguyen (Mid)	iOS/Android booking flow and digital passes; web client, operator dashboard, admin console

Area	Team members	Strengths
Mapping and location	Hannah Kim (Mid)	Map rendering, geolocation, search autocomplete
Infrastructure	Ethan Brooks (Mid)	Cloud hosting, CI/CD, scaling, monitoring and logging
Quality	Sofia Lindqvist (Mid)	Test planning, cross-platform testing, defect tracking, release readiness

Known gaps and how we cover them: senior expertise is concentrated in backend and payments, so mid-level developers pair with Carlos and Ibrahim (risks P1 and P4). The PO, BA, UX, SME and DevOps roles are part-time, so requests to them go through scheduled consults, and DevOps work is planned early because it peaks during setup and release.

2.5 Roles and Responsibilities

Role	Name	Allocation	Responsibilities
Project Manager	Isabelle Cruz	100%	Schedule (Gantt), risk register, sponsor updates, cross-team coordination, release go/no-go
Product Owner	Daniel Okafor	50%	Owns the Jira backlog, prioritizes tickets, accepts completed work, represents sponsor interests
Business Analyst	Priya Nair	75%	Maintains the SRS, writes use cases and acceptance criteria, traces requirements to tickets
UX Designer	Lucas Meyer	60%	Wireframes, user flows, Layer 2 mockups, operator dashboard layout
Subject Matter Expert	Rosa Delgado	15%	Parking operations and municipal regulations; validates pricing, occupancy and metered-parking logic
DevOps Engineer	Ethan Brooks	50%	Cloud hosting, CI/CD, load-balanced scaling (NFR3), monitoring and logging
Developer: Mobile	Aisha Rahman	100%	iOS/Android UI, booking flow, digital pass display
Developer: Web	Tom Nguyen	100%	Web client, operator dashboard, admin console
Developer: Backend/API	Carlos Vega	100%	Reservation, search and inventory services, data layer, role-based access control; technical lead
Developer: Mapping & Location	Hannah Kim	75%	Map rendering, geolocation, search autocomplete, mapping integration
Developer: Payment Integration	Ibrahim Yusuf	75%	Payment flow, refunds, receipts, PCI-DSS compliance checkpoints
QA Engineer	Sofia Lindqvist	100%	Test plans, cross-platform testing, defect tracking, release readiness

Task-level ownership (who is responsible, accountable, consulted and informed) is defined in the RACI matrix. Allocation is the share of a person's working time committed to Spotter.

2.6 Success Metrics

Metric	Target	How measured	Frequency
Schedule adherence	Milestones M1 to M5 met within one week of the Gantt baseline	Gantt v1.1 baseline vs actual	Every sprint review
Sprint commitment	At least 85% of committed tickets Done	Jira sprint report	Every sprint
Budget	Cumulative spend within the monthly cashflow; contingency used only with approval	Cost plan actuals vs plan	Monthly
Requirement coverage	100% of Layer 1 requirements accepted by the PO; FR10-FR12 mockups accepted	Ticket-to-requirement traceability	Every sprint, final UAT
Defect quality	No open critical or high defects at release	Jira bug tracker, QA weekly summary	Weekly
Performance	Search under 2 seconds (NFR2); occupancy data within 1 minute (NFR11)	Performance and load tests	Sprint 6, with spot checks
Security	PCI-DSS checklist fully passed; no cross-operator data exposure (NFR13)	Security review and access-control tests	Sprints 4 and 6
Risk health	No unmitigated risk scoring 16 or higher; register reviewed each sprint	Risk register	Every sprint
Team health	At least 80% of retrospective actions closed	Retrospective action log	Every sprint

2.7 Values

Value	What it means for us
Ownership	We own our tickets from start to Done and raise problems early rather than hoping they resolve.
Transparency	Status lives in Jira and the shared documents, so anyone can see where the work stands.
Quality and security first	We do not trade away testing, code review or payment security to save time.
Respect	We assume good intent, give feedback on the work and not the person, and respect part-time commitments.
Shared knowledge	We document and pair so no single person is a bottleneck.
Customer focus	We decide with drivers and parking operators in mind.

2.8 Team Norms

- **Availability:** core hours are 10:00 to 15:00 on weekdays (proposed), when everyone is reachable and meetings are held.
- **Jira is the source of truth:** update tickets the same day, and move a ticket to Done only when the Definition of Done is met.

- **Definition of Done:** acceptance criteria met, code reviewed by at least one teammate, automated tests passing in CI, verified by QA, and documentation updated.
- **Code review:** pull requests get a first review within one business day.
- **Responsiveness:** answer team chat within four working hours; use a call for anything urgent.
- **Blockers:** raise them the same day; if unresolved in 24 hours, escalate to the team lead and then the PM.
- **Decisions:** seek consensus first. If it is not reached, the PO decides priority, the technical lead decides technical approach, and the PM decides schedule. Once decided, everyone commits.
- **Conflict:** talk directly with the person first, then involve the team lead, then the PM.
- **Meetings:** start and end on time, have an agenda, and post notes and actions afterward.
- **Scope changes** go through the PO and PM, not directly to developers.
- **Document control:** SRS, WBS and Gantt changes are versioned and noted in the change log.

2.9 Communication Plan

Meeting	Cadence	Participants	Purpose and output
Daily standup	Daily, 15 minutes	Developers and QA; UX and DevOps as needed	Progress and blockers; Jira updated
Sprint planning	Every 2 weeks, start of sprint	All dev roles, PO, PM, BA	Commit sprint scope and goal
Cross-team sync	Weekly	Leads for Mobile, Web, Backend, Mapping, Payment, QA	Surface dependencies and blockers
Specialist syncs	Weekly; QA daily during test cycles	Mapping and Payment developers with Backend; QA with developers	Integration status, PCI-DSS checkpoints, defect triage
Sprint review and demo	Every 2 weeks, end of sprint	Team, PO, invited stakeholders	Demo finished work; PO accepts or carries over
Retrospective	Every 2 weeks, after the review	All team members	Improvements logged as Jira actions
Risk register review	Every sprint	PM and team leads	Update scores, owners and status
Sponsor update	Every 2 weeks, after the sprint review	PM, PO, sponsors	Status against the Gantt, risks and budget

Channels: Jira for ticket discussion, team chat for quick questions, email for formal or external communication, and calls for urgent matters. Communication with external stakeholders (operators, city departments, administrators, finance, payment providers, drivers and app users) follows the cadences in SRS section 9.2.

2.10 Coordination Tools

Purpose	Tool	Owner
Backlog, sprints and board	Jira (SPT project)	Product Owner, PM
Source control and code review	GitHub	Backend lead, all developers
Build, test and deploy	CI/CD pipeline with dev, staging and production environments	DevOps Engineer

Purpose	Tool	Owner
Team chat and meetings	MS Teams	PM
Schedule, risk and cost	Excel workbooks: Gantt v1.1, risk register, cost plan, RACI	PM
Requirements and project documents	SRS and WBS, versioned in Word/PDF	Business Analyst
Design and mockups	Figma	UX Designer
API documentation	OpenAPI specification	Backend developer
Defects and test management	Jira bug tracking and the weekly QA summary	QA Engineer
Logging and monitoring	Centralized logging and alerting (SPT-84, SPT-85)	DevOps Engineer

Tools marked proposed are not named in the SRS and can be swapped for the team's existing standards.

2.11 Team Agreement

By signing, each member agrees to the mission, goals, values and norms in this charter.

Name	Role	Signature	Date
Isabelle Cruz	Project Manager		
Daniel Okafor	Product Owner		
Priya Nair	Business Analyst		
Lucas Meyer	UX Designer		
Rosa Delgado	Subject Matter Expert		
Ethan Brooks	DevOps Engineer		
Aisha Rahman	Developer: Mobile		
Tom Nguyen	Developer: Web		
Carlos Vega	Developer: Backend/API		
Hannah Kim	Developer: Mapping & Location		
Ibrahim Yusuf	Developer: Payment Integration		
Sofia Lindqvist	QA Engineer		